Unitree Week 39

Presentation by Mario Ortiz

This Week

- Learning how to use the Unitree R1 Cameras
- Used Unitree UI to view cameras
- Used WebRTC to record videos

Investigating Camera Files

- Checked the Unitree R1 Library to search for any video, image, or camera files.

```
unitree@ubuntu:~$ find ~/unitree_sdk2 -type f | grep -Ei 'video|camera|image' | head -100
/home/unitree/unitree_sdk2/build/bin/go2_video_client
/home/unitree/unitree_sdk2/build/example/go2/CMakeFiles/go2_video_client.dir/DependInfo.cmake
/home/unitree/unitree_sdk2/build/example/go2/CMakeFiles/go2_video_client.dir/progress.make
/home/unitree/unitree_sdk2/build/example/go2/CMakeFiles/go2_video_client.dir/depend.make
/home/unitree/unitree_sdk2/build/example/go2/CMakeFiles/go2_video_client.dir/go2_video_client.cpp.o
/home/unitree/unitree_sdk2/build/example/go2/CMakeFiles/go2_video_client.dir/link.txt
/home/unitree/unitree_sdk2/build/example/go2/CMakeFiles/go2_video_client.dir/build.make
/home/unitree/unitree_sdk2/build/example/go2/CMakeFiles/go2_video_client.dir/depend.internal
/home/unitree/unitree_sdk2/build/example/go2/CMakeFiles/go2_video_client.dir/cmake_clean.cmake
/home/unitree/unitree_sdk2/build/example/go2/CMakeFiles/go2_video_client.dir/CXX.includecache
/home/unitree/unitree_sdk2/build/example/go2/CMakeFiles/go2_video_client.dir/flags.make
/home/unitree/unitree_sdk2/example/h1/doc/images/ankle.png
/home/unitree/unitree_sdk2/example/h1/doc/images/tracking.png
/home/unitree/unitree_sdk2/example/h1/doc/images/tracking_circle.png
/home/unitree/unitree_sdk2/example/go2/go2_video_client.cpp
/home/unitree/unitree_sdk2/include/unitree/robot/go2/video/video_error.hpp
/home/unitree/unitree_sdk2/include/unitree/robot/go2/video/video_api.hpp
/home/unitree/unitree_sdk2/include/unitree/robot/go2/video/video_client.hpp
/home/unitree/unitree_sdk2/include/unitree/robot/b2/back_video/back_video_api.hpp
/home/unitree/unitree_sdk2/include/unitree/robot/b2/back_video/back_video_error.hpp
/home/unitree/unitree_sdk2/include/unitree/robot/b2/back_video/back_video_client.hpp
/home/unitree/unitree_sdk2/include/unitree/robot/b2/front_video/front_video_client.hpp
/home/unitree/unitree_sdk2/include/unitree/robot/b2/front_video/front_video_api.hpp
/home/unitree/unitree_sdk2/include/unitree/robot/b2/front_video/front_video_error.hpp
/home/unitree/unitree_sdk2/include/unitree/idl/go2/Go2FrontVideoData_.hpp
```

Investigating Camera Files

- This command showed us where to find the R1's video client.
- We can request JPEG image data from Unitree's videohub services.
- It submits the request and saves JPEG images every 3 seconds.

```
unitree@ubuntu:~$ grep -RnsE 'VideoClient|GetImageSample|videohub' \
> ~/unitree_sdk2/include \
> ~/unitree_sdk2/example \
> 2>/dev/null | head -100
/home/unitree/unitree_sdk2/include/unitree/robot/go2/video/video_api.hpp:13:const std::string ROBOT_VIDEO_SERVICE_N
AME = "videohub";
/home/unitree/unitree_sdk2/include/unitree/robot/go2/video/video_client.hpp:13: * VideoClient
/home/unitree/unitree_sdk2/include/unitree/robot/go2/video/video_client.hpp:15:class VideoClient : public Client
/home/unitree/unitree_sdk2/include/unitree/robot/go2/video/video_client.hpp:18:    explicit VideoClient();
/home/unitree/unitree_sdk2/include/unitree/robot/go2/video/video_client.hpp:19:    ~VideoClient();
/home/unitree/unitree_sdk2/include/unitree/robot/go2/video/video_client.hpp:24:     * @brief GetImageSample
/home/unitree/unitree_sdk2/include/unitree/robot/go2/video/video_client.hpp:27:    int32_t GetImageSample(std::vect
or<uint8_t>&);
/home/unitree/unitree_sdk2/include/unitree/robot/b2/back_video/back_video_api.hpp:13:const std::string ROBOT_BACK_V
IDEO_SERVICE_NAME = "back_videohub";
/home/unitree/unitree_sdk2/include/unitree/robot/b2/back_video/back_video_client.hpp:13: * BackVideoClient
/home/unitree/unitree_sdk2/include/unitree/robot/b2/back_video/back_video_client.hpp:15:class BackVideoClient : pub
lic Client
/home/unitree/unitree_sdk2/include/unitree/robot/b2/back_video/back_video_client.hpp:18:    explicit BackVideoClien
t();
/home/unitree/unitree_sdk2/include/unitree/robot/b2/back_video/back_video_client.hpp:19:    ~BackVideoClient();
/home/unitree/unitree_sdk2/include/unitree/robot/b2/back_video/back_video_client.hpp:24:     * @brief GetImageSampl
e
/home/unitree/unitree_sdk2/include/unitree/robot/b2/back_video/back_video_client.hpp:27:    int32_t GetImageSample(
std::vector<uint8_t>&);
/home/unitree/unitree_sdk2/include/unitree/robot/b2/front_video/front_video_client.hpp:13: * FrontVideoClient
/home/unitree/unitree_sdk2/include/unitree/robot/b2/front_video/front_video_client.hpp:15:class FrontVideoClient :
public Client
/home/unitree/unitree_sdk2/include/unitree/robot/b2/front_video/front_video_client.hpp:18:    explicit FrontVideoCl
ient();
/home/unitree/unitree_sdk2/include/unitree/robot/b2/front_video/front_video_client.hpp:19:    ~FrontVideoClient();
/home/unitree/unitree_sdk2/include/unitree/robot/b2/front_video/front_video_client.hpp:24:     * @brief GetImageSam
ple
/home/unitree/unitree_sdk2/include/unitree/robot/b2/front_video/front_video_client.hpp:27:    int32_t GetImageSampl
e(std::vector<uint8_t>&);
/home/unitree/unitree_sdk2/include/unitree/robot/b2/front_video/front_video_api.hpp:13:const std::string ROBOT_FRON
T_VIDEO_SERVICE_NAME = "front_videohub";
/home/unitree/unitree_sdk2/example/go2/go2_video_client.cpp:13:    unitree::robot::go2::VideoClient video_client;
/home/unitree/unitree_sdk2/example/go2/go2_video_client.cpp:28:        ret = video_client.GetImageSample(image_samp
le);
unitree@ubuntu:~$
```

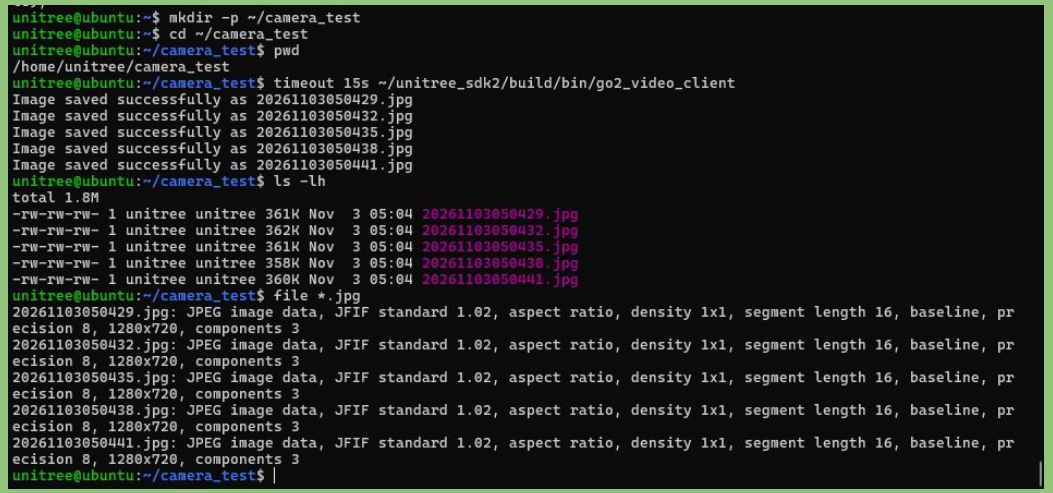
Command to take images on the R1

- Making a directory to test the camera feed.
- Successfully saved 5 images in 15s using video client.
- Specs of the images are:
  - 1280x720
  -

```
unitree@ubuntu:~$ mkdir -p ~/camera_test
unitree@ubuntu:~$ cd ~/camera_test
unitree@ubuntu:~/camera_test$ pwd
/home/unitree/camera_test
unitree@ubuntu:~/camera_test$ timeout 15s ~/unitree_sdk2/build/bin/go2_video_client
Image saved successfully as 20261103050429.jpg
Image saved successfully as 20261103050432.jpg
Image saved successfully as 20261103050435.jpg
Image saved successfully as 20261103050438.jpg
Image saved successfully as 20261103050441.jpg
unitree@ubuntu:~/camera_test$ ls -lh
total 1.8M
-rw-rw-rw- 1 unitree unitree 361K Nov  3 05:04 20261103050429.jpg
-rw-rw-rw- 1 unitree unitree 362K Nov  3 05:04 20261103050432.jpg
-rw-rw-rw- 1 unitree unitree 361K Nov  3 05:04 20261103050435.jpg
-rw-rw-rw- 1 unitree unitree 358K Nov  3 05:04 20261103050438.jpg
-rw-rw-rw- 1 unitree unitree 360K Nov  3 05:04 20261103050441.jpg
unitree@ubuntu:~/camera_test$ file *.jpg
20261103050429.jpg: JPEG image data, JFIF standard 1.02, aspect ratio, density 1x1, segment length 16, baseline, pr
ecision 8, 1280x720, components 3
20261103050432.jpg: JPEG image data, JFIF standard 1.02, aspect ratio, density 1x1, segment length 16, baseline, pr
ecision 8, 1280x720, components 3
20261103050435.jpg: JPEG image data, JFIF standard 1.02, aspect ratio, density 1x1, segment length 16, baseline, pr
ecision 8, 1280x720, components 3
20261103050438.jpg: JPEG image data, JFIF standard 1.02, aspect ratio, density 1x1, segment length 16, baseline, pr
ecision 8, 1280x720, components 3
20261103050441.jpg: JPEG image data, JFIF standard 1.02, aspect ratio, density 1x1, segment length 16, baseline, pr
ecision 8, 1280x720, components 3
unitree@ubuntu:~/camera_test$
```

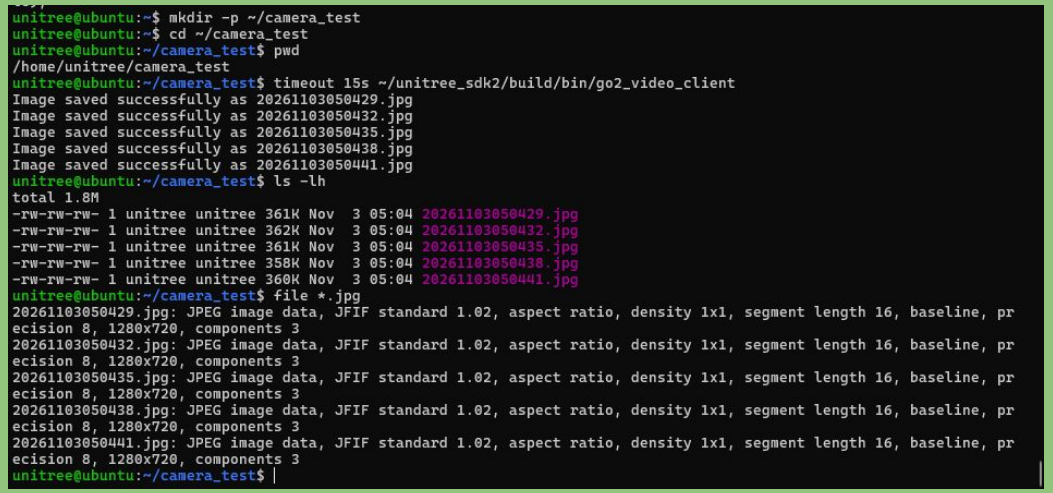
Transferring Images onto Laptop

- Transferred one of the images onto my computer and opened the file.
- We notice that though there are two cameras, the R1 photo is of one image, rather than two side-by-side images to indicate binocular view.



Findings Regarding Video

- The Jetson computer is not the host for the cameras, hence video control must be used via the robot itself.



Method to record on the R1: WebRTC

- WebRTC is a free open-source technology that allows web browsers and mobile apps to exchange audio, video, and other data in real time.
- WebRTC is a peer to peer streaming and control protocol used by the Unitree R1 to transmit live camera feeds, audio, and motion commands.

Obtaining the AES Key for WebRTC

- Installed Unitree UI
- Opened a local host to use remotely
- Signed in to Unitree Explore Account
- AES Key is provided once you sign in to the account that had been already linked to the R1.

Connecting R1 to the Unitree Ui (inputting AES-128 key)

- Command needed to open Unitree UI.

```
PS C:\Users\Mario> cd C:\Users\Mario\unitree_ui
PS C:\Users\Mario\unitree_ui> npm run dev

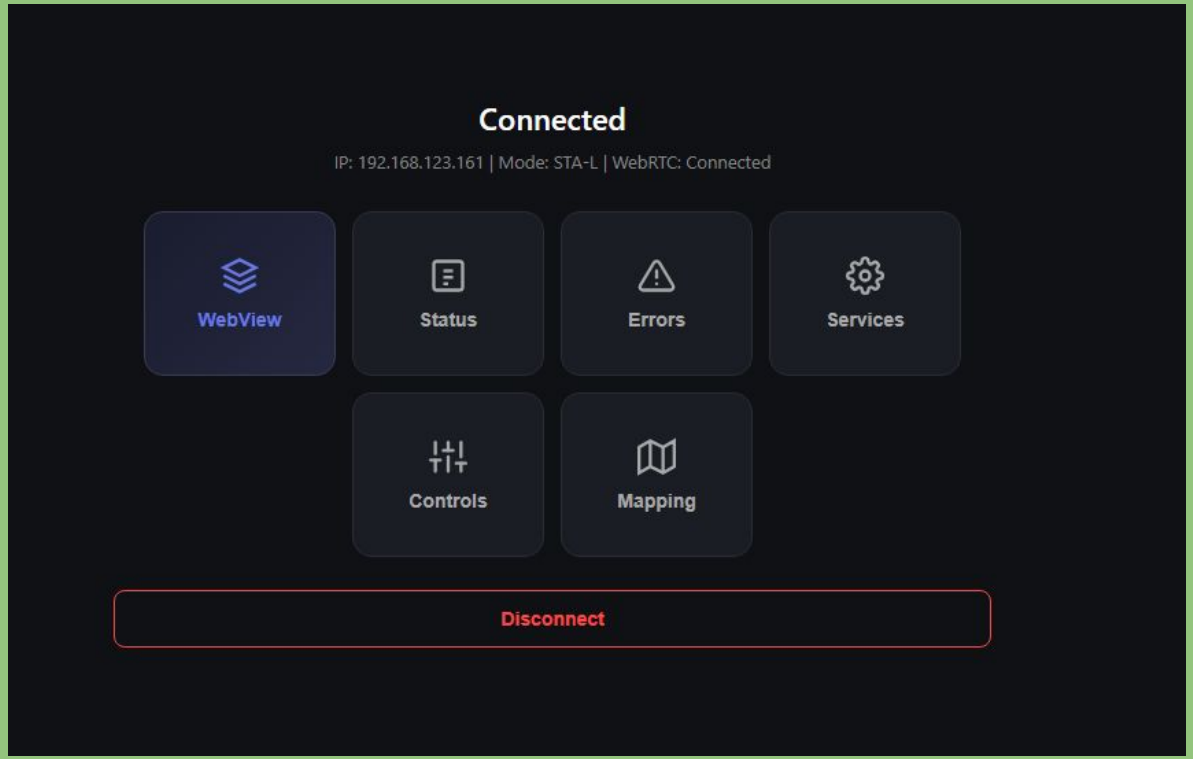
> unitree-ui@0.1.0 dev
> vite


  VITE v6.4.1  ready in 668 ms

  ->  Local:   http://localhost:5173/
  ->  Network: run `npm run dev:host` to expose on your LAN
  ➜  press h + enter to show help
```

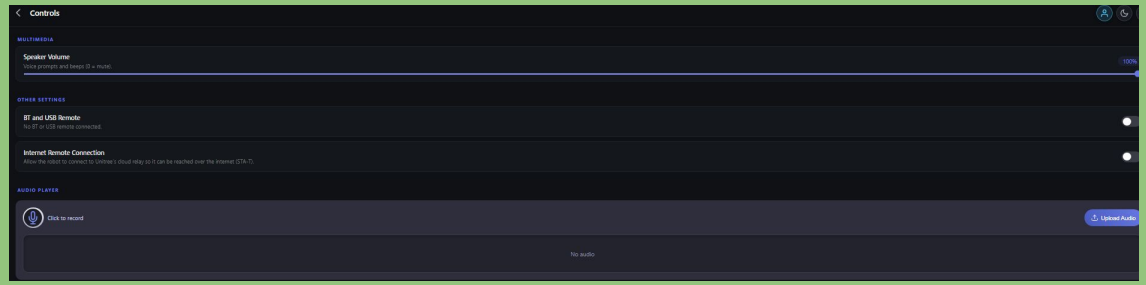
Connecting R1 to the Unitree Ui (inputting AES-128 key)

Overview of the Unitree UI.



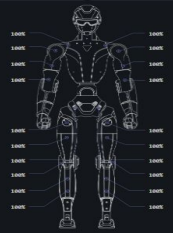
Connecting R1 to the Unitree Ui (inputting AES-128 key)

- Controls Section of the Unitree UI



MOTOR DATA

Comm Quality	Acc Loss	Position	Casing °C	Winding °C	Errors



Communication Quality	Good
Max Motor Temp	43°C

BODY IMU (TORSO)

Roll / Pitch / Yaw	R 0.00, P -0.06, Y -0.03
Temperature	79.0°C

CROTCH IMU (PELVIS)

Roll / Pitch / Yaw	R -0.01, P -0.05, Y -0.00
Temperature	79.0°C

Robot Status

SYSTEM

Family	R1
Mode	STA-L
IP Address	192.168.12.1
Serial Number	E39N4000QSGDN1GO
Hardware Version	1.0
Software Version	1.4.0
Machine Type	1
Motion Mode	ai
Connection Type	STA-L

BATTERY

Charge	51%
Voltage	33.3 V
Current	-2526 mA
Temperature	29°C
Charge Cycles	7
Pack Voltage	33.31 V
Cell Voltage	33.30 V
MOS Temp	29°C
BAT1 Temp	25°C
RES Temp	32°C

Recording Setup with WebRTC

- cd C:\Users\Mario\unitree_webrtc
- $env:UNITREE_ROBOT_IP="192.168.123.161"
- $secure = Read-Host "Enter R1 AES key" -AsSecureString
- $env:UNITREE_AES_128_KEY = [System.Net.NetworkCredential]::new("", $secure).Password
- Remove-Variable secure
- echo $env:UNITREE_ROBOT_IP
- .\.venv\Scripts\python.exe .\r1_record.py

Record.py (Program for recording with WebRTC)

```
PS C:\Users\Mario\unitree_webrtc> cat .\r1_record.py
import asyncio
import os
from datetime import datetime
from pathlib import Path

import cv2

from unitree_webrtc_connect import (
    UnitreeWebRTCConnection,
    WebRTCConnectionMethod,
)

ROBOT_IP = os.environ["UNITREE_ROBOT_IP"]
AES_KEY = os.environ["UNITREE_AES_128_KEY"]

output_dir = Path.home() / "Videos" / "R1"
output_dir.mkdir(parents=True, exist_ok=True)

filename = output_dir / (
    "R1_" + datetime.now().strftime("%Y-%m-%d_%H-%M-%S") + ".mp4"
)

writer = None
stop_event = asyncio.Event()


async def recv_camera_stream(track):
    global writer

    print("Video track received.")

    while not stop_event.is_set():
        frame = await track.recv()
        img = frame.to_ndarray(format="bgr24")

        if writer is None:
            height, width = img.shape[:2]

            print(f"Resolution: {width}x{height}")
            print(f"Recording to: {filename}")

            fourcc = cv2.VideoWriter_fourcc(*"mp4v")

            writer = cv2.VideoWriter(
                str(filename),
                fourcc,
                30.0,
                (width, height),
            )
```

```
                30.0,
                (width, height),
            )

        writer.write(img)

        cv2.imshow(
            "R1 Live Camera - press Q to stop recording",
            img
        )

        if cv2.waitKey(1) & 0xFF == ord("q"):
            stop_event.set()
            break


async def main():
    global writer

    print(f"Connecting to R1 at {ROBOT_IP}...")

    conn = UnitreeWebRTCConnection(
        WebRTCConnectionMethod.LocalSTA,
        ip=ROBOT_IP,
        aes_128_key=AES_KEY,
    )

    await conn.connect()

    print("WebRTC connected.")
    print("Starting R1 camera stream...")

    conn.video.add_track_callback(recv_camera_stream)
    conn.video.switchVideoChannel(True)

    try:
        await stop_event.wait()
    finally:
        conn.video.switchVideoChannel(False)

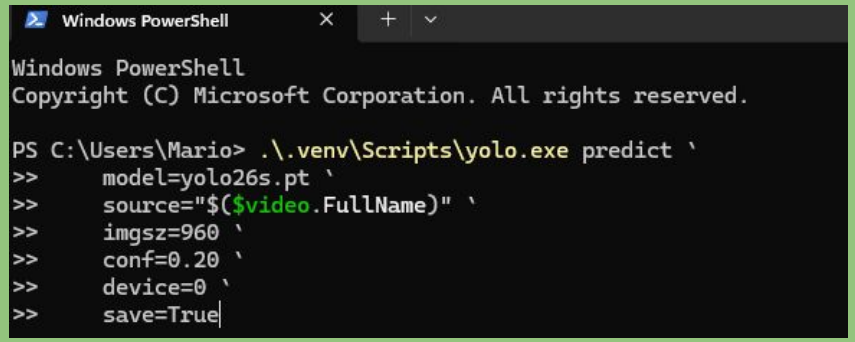
        if writer is not None:
            writer.release()

        cv2.destroyAllWindows()

        print()
        print("Recording finished.")
        print(filename)


asyncio.run(main())
```

Object Detection with Ultralytics and YOLO26



References

- https://github.com/unitreerobotics/unitree_sdk2
- https://github.com/unitreerobotics/unitree_sdk2/blob/main/example/go2/go2_video_client.cpp
- https://github.com/unitreerobotics/unitree_sdk2/blob/main/include/unitree/robot/go2/video/video_client.hpp
- https://github.com/unitreerobotics/unitree_sdk2_python/blob/master/example/go2/front_camera/camera_opencv.py
- https://github.com/unitreerobotics/unitree_sdk2_python/issues/102
- https://github.com/legion1581/unitree_webrtc_connect
- https://github.com/legion1581/unitree_ui/blob/main/README.md
- https://github.com/legion1581/unitree_ui/blob/main/docs/connection.md
- https://github.com/legion1581/unitree_ui/blob/main/docs/bluetooth-v3.md
- https://github.com/a-bissell/go2-webrtc-signaling/blob/main/disclosure_report.md
- https://github.com/mikeDK1/unitree-r1-gemini/blob/main/README.md
- https://docs.ultralytics.com/modes/predict
- https://docs.ultralytics.com/models/yolo26
- https://pytorch.org/get-started/locally/